SAMUEL J. TUSICK

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PROFESSIONAL SUMMARY

Software Engineering student building AI and automation systems that teams use every day. Automated an institute's daily operations and its AI avatar course production pipeline, co-designs an LLM-driven virtual patient platform with chat, voice, and avatar modes on a serverless AWS backend, and leads a student team building a machine learning payload for a high-power rocket. Have led three cross-functional teams. U.S. citizen, eligible to pursue a security clearance.

EDUCATION

Florida Gulf Coast University, B.S. Software Engineering *Fort Myers, FL | Expected May 2027*

- In-Major GPA: 3.83 | Dean's List (2023, 2024, 2025)
- Honors: Ruth M. and Cleveland L. Campbell Engineering Scholarship Endowed Fund, Kunkle Family Scholarship Endowed Fund
- Relevant Coursework: Machine Learning (in progress, Fall 2026), AI Agent Studio (in progress, Fall 2026), Software Architecture & Design (in progress, Fall 2026), Intro to Data Engineering, Data Structures & Algorithms, Software Engineering Fundamentals, Requirements Engineering & Analysis

TECHNICAL SKILLS

- **Languages:** Python, C++, C, SQL, JavaScript
- **Applied AI and Automation:** LLM API integration, prompt and skill design, agentic workflow automation, orchestration of multi-service AI systems
- **Machine Learning:** Neural networks and genetic algorithms from scratch (in progress); learning PyTorch and TensorFlow
- **Cloud:** AWS (API Gateway, Lambda, Aurora, Cognito, IAM, CDK), serverless architecture
- **Backend:** Flask, REST APIs, PostgreSQL, Supabase, SQLite
- **Frontend:** React, Vite, Tailwind CSS
- **Dev Tools:** Git, GitHub Actions, Selenium, Jira, Azure DevOps, pgAdmin, VS Code, CLion, OpenRocket

EXPERIENCE

FGCU Dendritic AI & Data Science Institute, Student Research Assistant *Mar 2025 to Present | On-site*

- Built an LLM-powered pipeline for the AIVA course production process: one skill turns professor PDFs and slide decks into narration scripts, and a second generates the AI avatar video and matching slide deck from each script, the selected avatar, and institute standards. Estimated 45% efficiency gain, used by the infrastructure and development teams.
- Built and maintain a reusable skill that generates the institute handbook, documenting every workflow and way of working, so the process takes a fraction of the time for each team member. Used institute-wide.
- Created a scheduled workflow that pulls Jira data every morning at 9 AM over HTTP and posts a daily briefing card to the team's Microsoft Teams channel, used by the whole institute.
- Built event-driven workflows that notify Teams channels when files are uploaded to monitored folders, supporting the digital media design and infrastructure teams.
- Coordinated logistics and content delivery for the 2026 FGCU Summer AI Academy (500+ participants, 150+ instructional hours) and built 14 curriculum-specific AI tutors for the Engineering Learning Hub.
- Evaluate emerging agentic AI frameworks for institute use.

Router-Level Security Detection, Subteam Lead and Backend Developer *Feb 2025 to Apr 2025 | On-site*

- Led backend architecture decisions and task coordination for a 5-member Eagle CyberNest Lab team, delivering a home-network security platform on schedule.
- Built Flask backend services with SSH and SQLite for real-time, centralized monitoring across three router platforms, and tested and deployed the router-communication scripts.

PROJECTS

MAIVA: AI Virtual Patient Training Platform | AWS, React, Vite, Tailwind, LLM API

- Co-own architecture, data modeling, and technology decisions, turning requirements from two clinical assistant professors into a platform where students practice clinical interviews with AI-simulated psychiatric patients. Instructors build scenarios from a DSM-5 patient profile and a persona; students run timed sessions and every transcript is saved for instructor review.
- Chat mode is in active testing with real mental health professionals. Voice mode is in active development, and avatar mode has its speech pipeline in place: text-to-speech viseme data drives facial animation on a 3D patient in Unreal Engine 5.
- Designed an orchestrator service that sits between the UI, speech-to-text, text-to-speech, and the LLM API for all three modes. It authenticates users through Cognito, streams the patient's reply sentence by sentence over a WebSocket, saves every turn to the database in real time, and rebuilds a session from the saved transcript after a restart. Each session starts from a master prompt filled with patient metadata from the selected scenario.
- Built the speech-to-text feature and implemented safety escalation handling that pauses the session clock so the student can resume or cancel. Refactored large parts of the React frontend.
- Run separate Local, Dev, and Test environments on a serverless AWS backend (API Gateway, Lambda, Aurora, Cognito, IAM, CDK).

Real-Time Landing Site Detection with Machine Learning | Python, Raspberry Pi, OpenRocket

- Team lead and software and hardware developer for a 4-person team selected as one of two winners of the FGCU Payload Challenge, earning funding and a flight on the club's high-power rocket.
- Designing a camera-only onboard payload where a segmentation model labels terrain as safe or unsafe and runs live on a Raspberry Pi alongside IMU, barometer, and GPS logging, with every frame and sensor reading saved for analysis.
- Chose to test during powered ascent rather than parachute descent to stress the model with flame, vibration, spin, and motion blur, the conditions a self-landing vehicle faces. Defined success targets of at least 2 predictions per second and 80% pixel accuracy, to be reported whether met or not.
- Built an OpenRocket flight simulation of the club rocket and ran an initial proof of concept with a pretrained model.

ML Self-Driving Car Simulator | Python, Pygame, Gymnasium

- Building a neural network and genetic algorithm trainer from scratch in Python, with no machine learning libraries, to teach a car agent to complete a track.
- Built the simulation: steering-based car physics, pixel-mask collision detection, five ray-cast distance sensors, a checkpoint system, and a Gymnasium-compatible environment.
- Established a measurable baseline before training: a random policy over 1,000 episodes finished the track 0 times, and only 2 episodes passed even one checkpoint.

Switch4Good Database System | PostgreSQL, SQL, Data Engineering

- Led development of a PostgreSQL database and schema for participation, projects, partnerships, and outcomes, supporting 100+ universities and 1,500+ students.

Pinterest Testing Suites | Java, Selenium, TestNG

- Led a 2-person team testing functionality, usability, and regression scenarios, contributing to all 46 test cases across 8 modules (personally authored 25) with a 97.8% pass rate.
- Diagnosed and resolved bot-detection rate limiting by restructuring login test timing.

NHL Stat Scraper | Python, Flask, Selenium, React, Vite

- Built a Selenium-based Flask scraper covering 15+ stat categories for every NHL player, exposed through REST APIs, with a React/Vite frontend for filterable stat views.

LEADERSHIP AND INVOLVEMENT

Deeptech Club, Payload Team Lead *June 2026 to Present*

Computer Science & Software Engineering Club, Member *Aug 2024 to Present*

Pinterest Testing Suites, Team Lead *Mar 2026 to May 2026*

FGCU AI Day, Panelist *Mar 2026*

Switch4Good Database System, Team Lead *Jan 2026 to May 2026*

Eagle CyberNest Lab, Subteam Lead (5-member team) *Feb 2025 to Apr 2025*